MECH 151: Technical Design – Prototyping

Prototype Design Proposal

Final Exam — Prototype Project (team submission)

Introduced Week 10 (Thursday, October 29, 2026) · Reviewed at the Week 11 design review/iteration workshop (Wednesday, November 4, 2026)

Team Name / Number: _____

Team Member 1: _____

Team Member 2 (if applicable): _____

Team Member 3 (if applicable): _____

Date: _____

What This Proposal Establishes

This proposal establishes your team, what you're proposing to build, and how your design will meet the Final Exam's two required elements: integrating at least 2 different bulk material types, and demonstrating at least 1 joining method between 2 parts. Proposals are reviewed during the Week 11 design review/iteration workshop (Wednesday, November 4, 2026) — expect feedback and a possible revision before you begin building. Together with the finished build, this project demonstrates Course Learning Objectives 4 (produce a component utilizing CAM software) and 5 (produce a prototype from multiple substrates).

Design Concept

What are you proposing to build? What problem does it solve or what function does it serve? Include a rough sketch or reference image if helpful.

Bulk Material Integration (required: at least 2 types)

List each bulk material type your design uses. A design must integrate at least 2 different types (e.g., thermoplastic + raw/processed stock).

Material #	Material Type	Used In / For	Why This Material
1			
2			

Add more rows if your design uses additional materials.

Joining Method (required: at least 1, between 2 parts)

Which joining method will you use (e.g., heat-set threaded insert, snap fit, press/interference fit, compliant/living hinge, threaded fastener)? Between which two parts? Why did you choose it over other options covered in Unit 2?

Manufacturing Plan

Which shop process(es) will you use to build this (additive/3D printing, subtractive/CNC, laser cutting)? What is your rough build timeline? (In-class prototype work time runs Weeks 13–16, Nov 19 – Dec 9; complete the safety/PPE session on Nov 18 before using shop equipment. The Final Exam is Wednesday, December 16, 2026.)

Proposal Review Notes

For instructor use during the Week 11 design review/iteration workshop.

Feedback:

Proposal Status (Approved / Revise & Resubmit): _____